Predicting vortex shedding behind a circular cylinder with physics-informed neural networks

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Abstract

A data-free physics-informed neural network (PINN) applied to the unsteady wake of a circular cylinder is known to lose the wake and fall towards the steady symmetric solution, which satisfies the governing equations exactly. This project reproduces that failure on purpose-generated lattice Boltzmann ground truth, validated at Re = 100 against Qu et al. (St within 0.04%, C_d 3.5% high), and measures what defeats it. Two confounds had inflated the project's own earlier results, and both are now controlled. One is how many slices of ground truth a run sees; the other is the phase of the shedding cycle at which it starts. With one slice and four start phases averaged over five seeds, a plain PINN keeps 31.1% of the reference wake amplitude. Causal time-marching raises that to 56.4%, although at equal training budget it adds only 7.7 points, less than the 17.5 that training longer adds on its own. Only with Fourier features and a Reynolds decomposition as well does the result stop depending on the start phase: 72.9% retained, at a velocity-field error of 0.031. Over five chained periods that configuration sheds 10.4% slow and ends about half a cycle out of phase. Re-anchoring it with 350 supervised points per shedding period, about 2% of the reference data that per-window snapshots use, holds the phase over all five. A few hundred supervised points per window close most of the remaining gap in amplitude. Across Re = 60–180 the three mechanisms keep 63–79% of the wake against 24–45% for the plain network. A recurrent sequence PINN is no substitute for marching: it keeps less of the wake than the plain network, marched or not. Adding the governing equations to a neural-operator forecast does not help on the validated data. On an earlier, harder dataset it extends a Fourier Neural Operator's useful horizon, but it is neutral for a U-Net and harmful for a convolutional GRU, so the benefit depends on the backbone.

1 Introduction and objective

Above Re $\approx$ 47 the wake behind a circular cylinder becomes unstable and sheds a Kármán vortex street at a frequency characterised by the Strouhal number St $= fD/U$. Predicting that shedding with a neural network is hard for three structural reasons:

1. **The steady solution remains a valid solution.** A symmetric, non-shedding field satisfies the Navier–Stokes equations and every boundary condition exactly. It is unstable physically, but a PINN loss measures residuals, not stability, so the steady field is an easily reachable minimum.

2. **Spectral bias.** Networks fit low-frequency content long before high-frequency content, which directly opposes representing an oscillatory wake.

3. **The initial condition is a measure-zero slice** of the space–time domain, and is therefore weakly weighted against the bulk PDE residual that fills it.

The documented consequence [1] is that a data-free PINN "behaves like a steady solver", while a data-driven one sheds only where data is present. Two points of scope. First, their cylinder case is at Re = 200, solved as a *single* time window spanning forty shedding periods. This project reproduces the phenomenon at Re = 100, and across Re = 60–180 in Section 7, from a baseline that solves one window of one period. That is a much easier setting than theirs, and the baseline still loses most of the wake. Second, they attribute the collapse to spectral bias, on the evidence of a Koopman analysis showing damped modes. The ablation below finds that attacking spectral bias directly does not, on its own, keep more of the wake. Enforcing causality does, and does most in combination with the other two mechanisms.

The objective is to reproduce that failure on validated ground truth of our own, and then to establish *which* mechanism defeats it, with an ablation controlled well enough that the answer is attributable. The mechanism that works, causal time-marching, is not new in itself: it is the sequence-to-sequence formulation of Krishnapriyan et al. [4], demonstrated there on one-dimensional convection, reaction and reaction–diffusion problems, and a hard-window relative of the causal weighting of Wang et al. [5]. What is measured here is how much it contributes on two-dimensional wake shedding without data, separated from the extra training a marched run receives, and how it interacts with the other two mechanisms. Chuang and Barba [1] report that some form of data supervision is necessary to sustain shedding at all. So does Shah [3], more recently, who ablates six techniques on a confined cylinder benchmark and finds that only supervised combinations escape the collapse. Neither tests hard time-window marching. The answer found here sits between: a data-free marched chain does sustain a wake, at three-quarters of its amplitude, but only with all three mechanisms together, and over five periods it drifts about half a cycle out of phase with the reference. Holding the phase does take data, but little: 350 supervised points per shedding period.

A second thread connects the work to its reference paper. Renn et al. [2] forecast subcritical cylinder wakes with a Fourier Neural Operator (FNO) that is *purely data-driven*; there is no physics term anywhere in it. Section 9 tests whether adding physics to such an operator improves forecast stability, for the FNO and for two other common forecasting backbones. Section 8 asks the corresponding question of the PINN: whether a sequence architecture proposed against PINN failure modes does the job of marching.

Reynolds number. The ablation and the accuracy study are at Re = 100: laminar and two-dimensional. Renn et al.'s Re = 240–3060 is three-dimensional, where a 2-D model would be solving the wrong physics. Section 7 repeats the central comparison across Re = 60–180, the upper limit set by the three-dimensional transition at $\mathrm{Re}_c = 188.5 \pm 1.0$ [16].

Software. The work builds on `underPINN`, a JAX PINN framework, which supplies the MLP and Fourier-feature networks, the FNO, the geometry primitives and a steady Navier–Stokes residual. The unsteady operator (Section 3), the lattice Boltzmann solver, the ablation machinery, the operator-forecasting code and the sequence, U-Net and ConvGRU networks of Sections 8 and 9 are contributions of this project.

2 Ground truth by lattice Boltzmann

No experimental data was available, so the reference wake is generated. The solver (`src/cfd/lbm_cylinder.py`) is a D2Q9 two-relaxation-time (TRT) lattice Boltzmann method written in JAX.

Solver choices.

- **TRT collision.** A single-relaxation-time (BGK) scheme places the bounce-back wall at a viscosity-dependent position, so the cylinder's effective diameter drifts with Re. TRT relaxes the symmetric and antisymmetric parts separately, allowing the magic parameter $\Lambda = (\tau^+ - 1/2)(\tau^- - 1/2)$ to be pinned at 3/16, which fixes the wall location.
- **Half-way bounce-back** for the no-slip cylinder, placing the wall midway between nodes, and **momentum exchange** for the surface force.
- **Specular (free-slip) top and bottom walls**, enforcing $v = 0$ exactly and matching the symmetry condition used on the PINN side.
- **An absorbing layer** of $8D$ at each free-slip wall, relaxing the populations toward the free-stream equilibrium at a peak rate of 0.05 per step on a cubic ramp, with τ^- recomputed pointwise so that Λ stays exactly 3/16. Section 2.1 explains why it is needed.
- **The inlet $25D$ ahead of the cylinder**, in a domain $60D$ long.

Shedding is triggered by a brief transverse inlet perturbation; without it the symmetric base flow persists indefinitely (the same degeneracy that the PINN later falls into, appearing here in the reference solver).

Production dataset. Re = 100, $D = 24$ lattice units, Ma = 0.087, grid 1440×1200, blockage $D/H = 2\%$. The saved record is 641 snapshots covering 20 shedding periods, on a 241×121 grid at $\Delta x = 0.083D$ and $\Delta t^* = 0.19$, chosen near the reference paper's PIV resolution so that the later comparison is like-for-like.

Table 1: Validation of the ground truth at Re = 100. Benchmarks are Qu et al. [14]: the range is their compilation of eight independent large-domain studies, and the errors are taken against their own values, $C_d = 1.317$, St = 0.1649 and $C_l' = 0.2224$.

quantity	computed	benchmark [14]	error
C_d (mean drag coefficient)	1.3634	1.318–1.336	+3.5%
St (Strouhal number)	0.1648	0.164–0.166	−0.04%
C_l' (r.m.s. lift)	0.2362	0.222–0.235	+6.2%

St is measured from wake vorticity, which cannot see irrotational sound; the lift and a single wake probe give the same value to within 0.002%. The remaining C_d offset is resolution, quantified below: at $D = 32$ the same geometry gives +2.3%.

2.1 How the ground truth was validated

An earlier version of this dataset put the inlet $8D$ ahead of the cylinder and had no absorbing layer. It read St = 0.1595 (−3.3%) and $C_d = 1.3798$, and was reported as agreeing with a benchmark of 1.37–1.38 to 0.3%; that benchmark was uncited and closer to the Re = 200 value. Three problems were found and fixed, each by a controlled test (Figure 1).

The domain is a duct, and the duct resonates. Free-slip walls reflect sound perfectly and lattice Boltzmann carries real sound, so the domain supports a transverse standing half-wave at $c_s/2H$ (in flow units $\mathrm{St}_a = (c_s/2H)(D/U) = 0.1155$). Given time it captures the wake itself: with no absorbing layer, wake vorticity sheds at St = 0.1600 over 20 recorded periods and at 0.1132 over 38, on the resonance. A viscosity ramp, the textbook absorbing layer, does nothing here (0.1131): viscous absorption scales as $\nu\omega^2$ and this mode's wavelength is twice the domain height, so an $8D$ layer at twenty times the viscosity removes about 0.04% of the amplitude per

pass. Relaxing toward the free stream damps at a rate set per step, independent of frequency, and releases the wake (0.1694 over the same 38 periods). The result does not depend on the layer: across a twofold change in its width and a fivefold change in its rate, C_d moves by 0.1% and St by 0.2%, and 38 and 120 recorded periods agree to four figures.

Blevins [18] showed experimentally that a transverse acoustic wave shifts cylinder shedding to its own frequency through the acoustic velocity rather than the pressure, and Hong et al. [17] model the coupling for a cylinder in a duct, placing the lock-in band at $\mathrm{St}/\mathrm{St}_a \leq 1.14$. At $\mathrm{Re} = 100$ the ratio here is 1.43, well outside that band, and the wake still locks given enough time, plausibly because lossless walls sustain the mode at separations a physical duct could not. The practical consequence was that no dataset could be recorded for long enough to average force coefficients over the 80-odd cycles the literature uses.

Inlet distance and blockage. With the absorbing layer in place and the grid converged, crossing domain height ($H = 50D$, $120D$) with inlet distance ($8D$, $25D$) separates the two geometric suspects. Moving the inlet from $8D$ to $25D$ ahead of the cylinder lowers C_d by 4.7%, from +7.4% to +2.3% against Qu et al., and takes St from +3.0% to +0.3%; raising the height from $50D$ to $120D$ moves C_d by 0.36%, and by 0.07% once the inlet is far enough upstream. Posdziech and Grundmann, as reported by Qu et al. [14], require more than $20D$. Blockage was never the dominant error.

Resolution. At fixed geometry, $C_d = 1.4334$, 1.4139 and 1.4147 at $D = 24$, 32 and 48: converged at $D = 32$, with $D = 24$ about 1.4% high in C_d and within 0.3% in St. The production dataset keeps $D = 24$ so that its snapshot grid matches the PIV-like resolution the networks are trained on; the price is the C_d offset in Table 1.

Residual divergence. Lattice Boltzmann is weakly compressible, so $\nabla \cdot \mathbf{u}$ is never zero. Over the snapshot grid, excluding the cylinder and a two-cell halo, its r.m.s. is 2.46% of $|\nabla \mathbf{u}|$, against $\mathrm{Ma}^2 = 0.75\%$. A spectral decomposition separates two parts. The time-mean part, the steady compressibility of the mean flow, is unchanged from the earlier dataset. The fluctuating part is broadband and strongest in the far wake; it is a property of the longer domain, not of the absorbing layer (the same geometry without the layer carries the same level), and the coherent duct mode that dominated it before is gone.

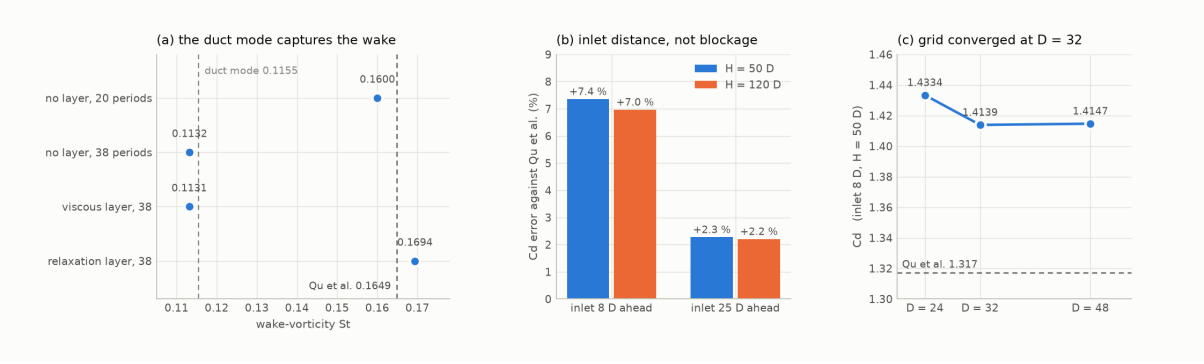


Figure 1: How the ground truth was validated. (a) Wake-vorticity St: without an absorbing layer the wake locks onto the transverse duct mode given enough time; a viscous layer does not prevent it; relaxation toward the free stream does. (b) C_d error at $D = 32$ with the layer: the inlet distance was a larger geometric error than the domain height. (c) Grid convergence at the original inlet distance.

Measuring St. An early estimate of St taken from an FFT peak gave a fraction-of-a-percent agreement that was pure luck: over a record of this length a single FFT bin is wider than the

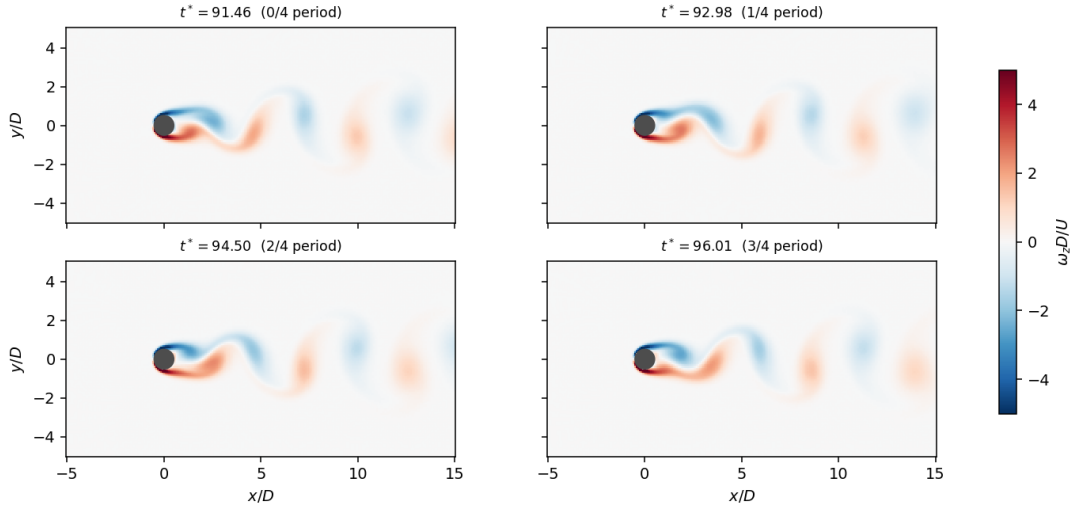


Figure 2: The reference Kármán street at quarter-period intervals, the start of the record. These four instants are the four start phases of Section 4.

discrepancy being measured. The St values reported here are measured from wake vorticity averaged over many points, cross-checked against a single wake probe and the lift; on the production dataset the three agree to 0.002%, where on the earlier one they disagreed by about 1% because the lift carried the duct mode.

3 An unsteady Navier–Stokes operator

`underPINN` ships only a steady incompressible residual, which cannot represent shedding at all. This project supplies the unsteady operator (`src/pinn/ns2d_unsteady.py`), conforming to the framework's `BasePDE` contract so it composes with the existing solvers, samplers and networks. Non-dimensionalised on D and U,

$$u_x + v_y = 0, \tag{1}$$

$$u_t + uu_x + vu_y + p_x - \tfrac{1}{\mathrm{Re}}(u_{xx} + u_{yy}) = 0, \tag{2}$$

$$v_t + uv_x + vv_y + p_y - \tfrac{1}{\mathrm{Re}}(v_{xx} + v_{yy}) = 0, \tag{3}$$

with the network mapping $(x, y, t) \mapsto (u, v, p)$.

Two derivative backends are implemented: a literal `jax.hessian` reference, and a forward-over-forward JVP path computing only the six derivative groups the residual needs, where a full Hessian would compute 27.

Table 2: Verification of the unsteady operator. All three checks pass at machine precision.

check	result
annihilates the Taylor–Green vortex (an exact NS solution)	1.6×10^{-16}
fast JVP backend versus literal Hessian	2.8×10^{-17}
steady limit versus `underPINN`'s own `NavierStokesPDE`	5.6×10^{-17}
speed-up of the fast backend	$\approx 2.5\times$ (load dependent)

The third check compares the new operator, in its steady limit, with the framework's own steady residual, so it tests the new code against independent code that was already in use.

4 Reproducing the collapse

A plain multilayer perceptron $[3, 128^{\times 6}, 3]$ is trained on physics, boundary and initial conditions with *no data*, over a single time window of one shedding period, for 20 000 epochs. Its initial condition is the reference field at the start of the window, the one piece of ground truth that any forward solve needs.

Shedding retained is this project's own metric: the ratio of transverse-velocity r.m.s. at a fixed wake probe between the PINN and the LBM reference over the run. Near 0% means collapse to steady flow; 100% means the wake amplitude matches.

Table 3: The plain PINN, by the phase of the shedding cycle at which it is started. Quarter-period steps, five seeds each, seed means.

start phase	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	mean
shedding retained	43.2%	20.3%	42.1%	19.0%	**31.1%**

The plain PINN keeps between 19% and 43% of the reference amplitude, with a field error of 0.17: it does not reproduce the wake, though within one period it does not collapse fully onto the steady solution either. Which of those it looks like depends on *when it starts*. Retention alternates every quarter period, so it repeats every half cycle, the half period over which the transverse velocity at the probe changes sign. Released at a peak of the wake signal the network carries more of it through the window than released at a zero crossing.

An earlier version of this report quoted "10.6% of the wake" from a single starting instant; on that dataset the same configuration gives 9.1%, 25.6%, 13.1% and 34.9% at the four phases. A comparison between configurations at one starting instant is a comparison at that phase only, so every retention figure below is either averaged over the four phases or labelled with its phase.

Longer training. Trained for 80 000 epochs instead of 20 000 (the budget a four-window marched run spends in Section 5), the same network keeps 48.6% on average, at a field error of 0.115. Training longer helps, by 17.5 points phase-averaged, but the half-cycle swing survives it: 62.6% and 61.2% at phases 0 and $\frac{1}{2}$, 34.3% and 36.5% at the other two.

5 Defeating the collapse

Three mechanisms are implemented as switches on one trainer, so that the ablation is controlled: every configuration shares the sampler, the optimiser, the schedule and the evaluation.

Table 4: The ablation matrix: what each configuration attacks.

configuration	mechanism	obstacle attacked
`baseline`	—	(control)
`fourier`	random Fourier features	spectral bias
`decompose`	Reynolds decomposition $u = u_0 + u'$	conditioning, dynamic range
`march`	causal time-marching	weak initial condition, causality
`all`	all three	—
`all+data`	plus 4 000 supervised points per window	—

There are three implementation details. First, the Reynolds decomposition requires a *differentiable* mean field, because the momentum residual contains ∇u_0 and $\nabla^2 u_0$; bilinear interpolation of a grid gives piecewise-constant first derivatives and identically zero second derivatives. The mean is therefore fitted by a separate frozen network (relative $L_2 = 2.2 \times 10^{-3}$). Second,

that mean is obtained by *regression on the measured mean*, not by a steady PINN solve, because the time mean of a shedding flow does not satisfy the steady equations; it carries the divergence of the Reynolds stresses.

Third, the marching chain. A marched run solves four windows of a quarter period in sequence. The first takes the reference initial condition, like every other configuration; each later window takes its initial condition from the *previous window's network* at the shared instant, so the whole run touches ground truth exactly once. An earlier version of this project instead gave every window a fresh reference snapshot, so a marched run used four ground-truth slices spread across the period against one for the single-window runs it was compared with. The effect of that is measured below.

5.1 Result

Table 5: The ablation at Re = 100, start phase 0: 20 000 epochs per window, two seeds, seed means, one ground-truth slice per run.

configuration	shedding retained	relative L_2
`baseline`	41.1%	0.173
`fourier`	36.9%	0.149
`decompose`	36.1%	0.086
`march`	72.2%	0.156
`all`	77.3%	0.052
`all+data`	99.1%	**0.006**

Table 6: The three configurations the conclusion rests on, at all four start phases (five seeds each). `all` uses $\sigma = 8$.

configuration	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	mean	relative L_2
`baseline`	43.2%	20.3%	42.1%	19.0%	31.1%	0.166
`march`	70.6%	40.4%	73.8%	40.7%	56.4%	0.156
`all`	74.9%	70.2%	76.6%	70.0%	**72.9%**	**0.031**

Neither spectral bias nor conditioning is the binding constraint on its own: Fourier features and the decomposition each leave retention slightly below the plain PINN's (37% and 36% against 41% at phase 0), though the decomposition halves the field error. Causal time-marching roughly doubles the retained wake at every start phase, not only on average, though with four times the training (see *Equal compute* below); but on its own it inherits the plain PINN's half-cycle swing, between 40% and 74%. Only with all three mechanisms does the result stop depending on when the run starts: 70–77% at every phase, at a field error of 0.031, a fifth of either of the others. Over the five seeds the phase means carry 95% intervals of ±1.5, ±0.9 and ±4.6 points for `baseline`, `march` and `all`: `all` varies most from seed to seed, but nowhere near enough to close either gap. No single mechanism is sufficient; the three are complementary, which agrees with the interaction effects Shah [3] reports.

What the extra ground truth was worth. The same marched configurations with a fresh reference snapshot at every window, as the earlier version ran them, retain 93.0% (`march`) and 94.5% (`all`) at phase 0, against 72.2% and 77.3% chained. About twenty points of the earlier headline were three extra slices of the answer.

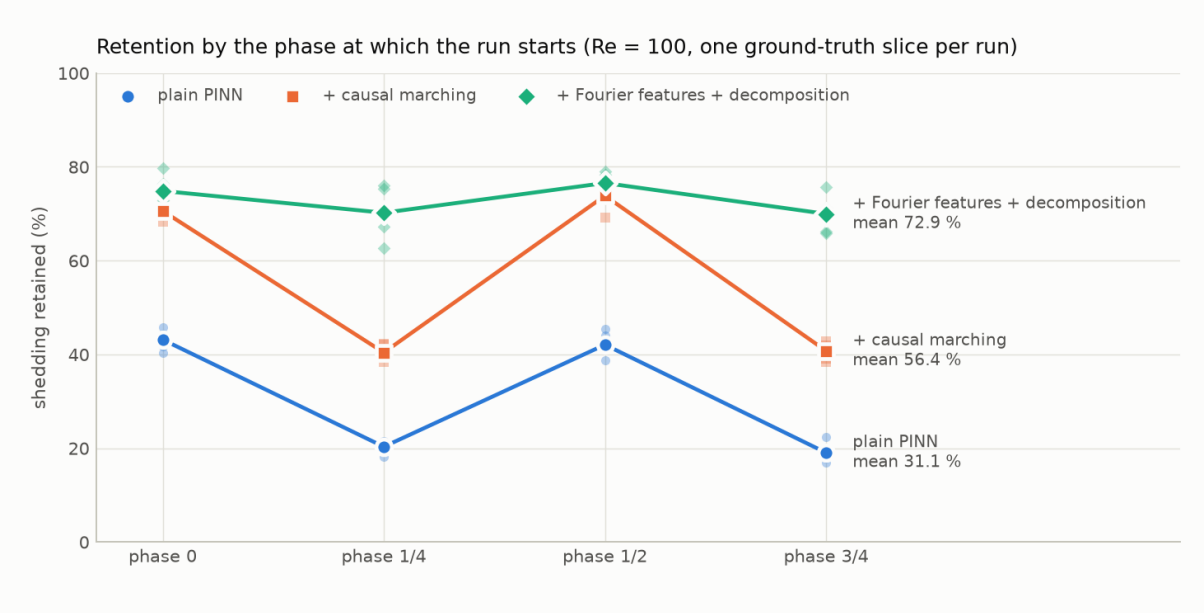


Figure 3: Shedding retained by start phase, every run plotted. Marching lifts the wake at every phase but keeps the plain PINN's half-cycle swing; with Fourier features and the Reynolds decomposition added, the result no longer depends on when the run starts.

Equal compute. A marched run trains four windows of 20 000 epochs, four times what a single-window run gets, so part of its advantage could be training alone. Each marched configuration was therefore paired with a single-window run given 80 000 epochs, at every start phase and seed.

Table 7: Marching against single-window runs of equal training budget (80 000 epochs), averaged over four start phases and five seeds. Gains are paired by phase and seed.

comparison	retained	retained	gain (points)	paired range
`march` vs `baseline`, 80 000	56.4%	48.6%	+7.7	−2.2 to +15.8
`all` vs Fourier + decomposition, 80 000	72.9%	38.7%	+34.1	+20.0 to +55.2

Alone, marching wins 19 of the 20 pairings and loses one by 2.2 points, and it wins by less than the first table suggests. Of its gain over the plain PINN, training longer accounts for 17.5 points and causality for 7.7. With Fourier features and the decomposition in place the picture changes. Those two mechanisms on one window keep 38.7% at 80 000 epochs, less than the plain PINN given the same budget, and they still swing with start phase; adding the marching chain lifts them by 34.1 points, in all 20 pairings, and removes the swing. Causality is worth more than four times as much in the combination as alone. Over the five seeds the three paired gains carry 95% intervals of ±0.7 points (training longer), ±1.9 (causality alone) and ±9.0 (causality in the combination). The last is wide because its one-window control varies more between seeds than any other configuration, at 38.7 ± 9.5%.

5.2 Hyperparameters

The Fourier bandwidth no longer changes how much wake survives, but it lowers the field error by forty per cent once the other mechanisms are in place, and does little without them. A heavier initial-condition weight buys nothing.

Table 8: Knobs, start phase 0, two seeds each.

change	applied to	shedding retained	relative L_2
$\sigma = 8$ (from 2)	`all`	77.3% → 76.4%	0.052 → **0.031**
$\sigma = 8$	`fourier`	36.9% → 37.6%	0.149 → 0.136
$w_{ic} = 100$ (from 10)	`all`	77.3% → 74.7%	0.052 → 0.065

5.3 How much supervision is needed

Table 9: Supervised points per window on `all`, start phase 0, two seeds, seed means.

data points per window	shedding retained	relative L_2
0	77.3%	0.052
100	87.4%	0.020
500	96.9%	0.010
2000	98.1%	0.007
4000	99.1%	0.006

A few hundred supervised points per window close most of the remaining gap in amplitude, and the field error keeps falling, ninefold by 4000 points.

6 Shedding accuracy

Shedding retained is an amplitude ratio: it answers "is there a wake", which is the right question for the collapse and the wrong one for prediction accuracy. A network could hold the right amplitude at the wrong frequency, or drift out of phase, and the ablation table would not notice. The best data-free configuration (`all`, $\sigma = 8$) was therefore compared against the reference probe signal directly, over one shedding period and, marched over twenty chained windows, over five.

Table 10: Prediction accuracy of `all` ($\sigma = 8$) at $\mathrm{Re} = 100$, one ground-truth slice per run.

quantity	1 period (4 windows)	5 periods (20 windows)
shedding frequency St	—	0.1476 (−10.4%)
reference St over the same span	—	0.1648
zero crossings available	2	9
wake amplitude	73.5%	63.8%
phase lag	0.1°	—
phase error by the end	—	≈0.5 period
velocity-field error	0.034	0.156

Over one period the network holds the phase of the reference to a tenth of a degree, at 73.5% of its amplitude and a field error of 0.034. No frequency is quoted for that run: two zero crossings are not enough to measure one, and over the same window the estimator reads even the reference at 0.139 against the dataset's 0.1648.

Over five periods the network keeps shedding (nine zero crossings and a recognisable Kármán street, Figure 4) but at a frequency 10.4% below the reference. The phase error that accumulates from that is about half a cycle by the end of the run, which is what the probe trace shows: the two signals start together and finish nearly in antiphase. Amplitude decays to 63.8% and the

field error grows to 0.156. A cross-correlation lag is not quoted for this run, because with the frequencies a tenth apart it is ambiguous modulo one period.

An earlier version of this report found the opposite (St within 0.1% over five periods and a negligible drift) and concluded that the marching chain does not accumulate error. That run gave all twenty windows a fresh reference snapshot, so the phase was re-anchored to the truth at every handoff. A chain that carries its own state forward does not hold phase over five periods. Data-free marching sustains a wake, but on this evidence it does not forecast one.

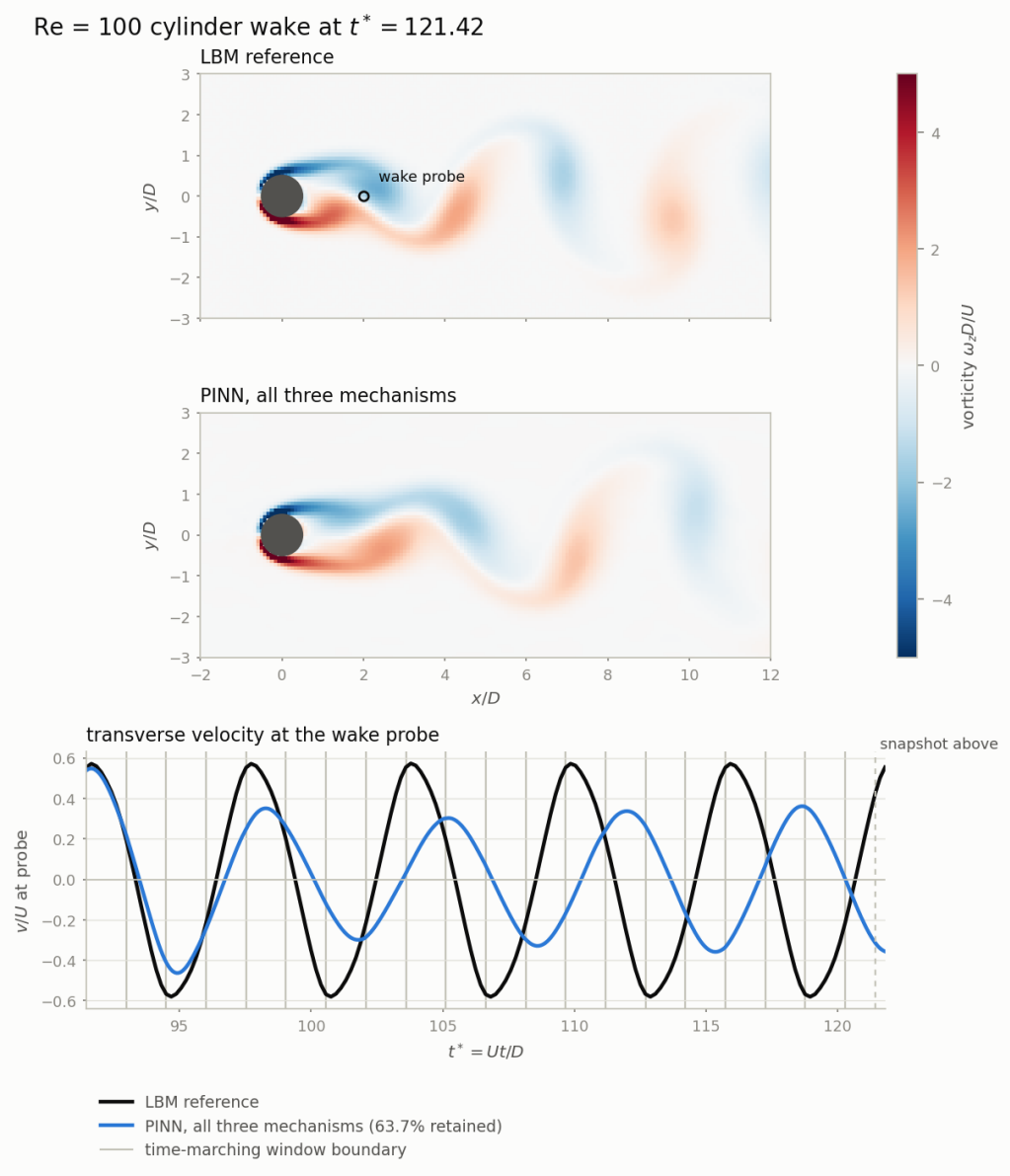


Figure 4: The data-free PINN (`all`, $\sigma = 8$) marched over five shedding periods on one ground-truth slice, against the reference. Top: wake vorticity at the end of the run. Bottom: transverse velocity at the wake probe; grey lines are window boundaries. The network tracks the reference for about a period, then sheds slower and weaker, and ends nearly half a cycle behind.

Visible artefact. The probe trace has small kinks at window handoffs, where continuity between windows is enforced only through the initial-condition loss. Roy and Castonguay [13] enforce it exactly through a solution ansatz; applying that here is listed in Section 11.

Sparse re-anchoring. The earlier run held the phase with a fresh snapshot at every window: $20 \times 4\,000 = 80\,000$ reference points. Table 11 asks how few will do. The five-period run is repeated with N supervised (u, v) samples, drawn across the window's own time span, given to every K-th window (windows $0, K, 2K, \ldots$) and to no other. Initial conditions stay chained throughout, so after the first slice the anchor points are the only ground truth a run sees. The phase is scored as the cross-correlation lag within the last reference period, which is unambiguous while the drift per period stays under half a cycle; for the data-free run it reads 0.37 of a period, a little below the end-point estimate of Table 10 because it averages over the final period.

Table 11: Sparse re-anchoring of the five-period run (`all`, $\sigma = 8$, 20 chained quarter-period windows). N supervised points go to every K-th window. Two seeds per row, each listed; the lag is in shedding periods, over the last period, positive behind the reference.

N	K	points per period	total points	St error (%)	lag at the end
—	—	0	0	−10.4, −10.2	0.369, 0.345
100	8	60	300	−8.8, −9.7	0.327, 0.328
100	4	100	500	−7.6, −9.1	0.220, 0.318
100	2	200	1 000	−4.6, −3.3	0.120, 0.100
125	2	250	1 250	−1.7, −5.7	0.070, 0.185
150	2	300	1 500	−1.5, −1.3	0.053, 0.042
75	1	300	1 500	−1.5, −2.6	0.037, 0.080
300	4	300	1 500	−3.3, −3.1	0.088, 0.107
175	2	350	1 750	−0.2, −0.7	0.002, 0.002
100	1	400	2 000	−0.3, −0.2	0.001, 0.001
500	4	500	2 500	−0.3, +0.5	0.002, 0.003

Up to 300 points per period the anchors slow the drift but do not stop it: 200 per period still leaves the run 3–5% slow and a tenth of a period behind. At 350 per period and above, both seeds hold the phase to within 0.003 of a period over all five periods and St to within 0.7%, on as few as 1 750 points (about 2% of what the per-window snapshots used). The threshold therefore lies between 300 and 350 points per period. Near it the schedule matters a little: at 300 per period, one anchor per period is the worst of the three schedules on both seeds, while one in every window and one every half period overlap within the seed spread. Amplitude rises with the supervision too, from 63–64% data-free to 74–87% where the phase locks.

7 Other Reynolds numbers

Everything above is at $\mathrm{Re} = 100$. The comparison was repeated at five more Reynolds numbers spanning the two-dimensional shedding range, with fresh ground truth generated at each by the same solver, geometry and absorbing layer as the production dataset. The marching window is set to a quarter of each dataset's *measured* shedding period, so all configurations span exactly one period whatever the frequency; otherwise the comparison would not be like-for-like across Re. Every run sees one slice of ground truth, as in Section 5, and `all` uses $\sigma = 8$.

The result generalises. At every Reynolds number tested, the plain network keeps 24–45% of the wake and the three mechanisms together 63–79%, and they cut the field error from about 0.17 to 0.028–0.040. The ground truth holds up across the range as well: every dataset's St is within 0.6% of Qu et al. So the result holds across the two-dimensional shedding range, and is not tied to one dataset. $\mathrm{Re} = 100$ reproduces the main ablation's first start phase, whose two configurations it shares: 38.8 and 41.5% for `baseline` here against 40.4 and 41.7% there.

Table 12: The plain network and all three mechanisms across the shedding range, data-free. Shedding retained for each of two seeds; field error is the two-seed mean. St is each dataset's own wake-vorticity value against Qu et al. [14], Table 3.

Re	St vs Qu et al.	`baseline`	`all`	relative L_2 `baseline`	relative L_2 `all`
60	+0.51%	29.8, 33.5%	**62.8, 69.1%**	0.163	**0.028**
80	+0.60%	40.2, 44.6%	**75.5, 76.9%**	0.173	**0.028**
100	−0.04%	38.8, 41.5%	**71.8, 78.5%**	0.173	**0.031**
120	+0.03%	23.9, 24.1%	**71.5, 78.6%**	0.170	**0.036**
150	−0.28%	38.0, 41.4%	**69.7, 75.1%**	0.174	**0.039**
180	−0.45%	24.1, 27.7%	**77.5, 76.6%**	0.171	**0.040**

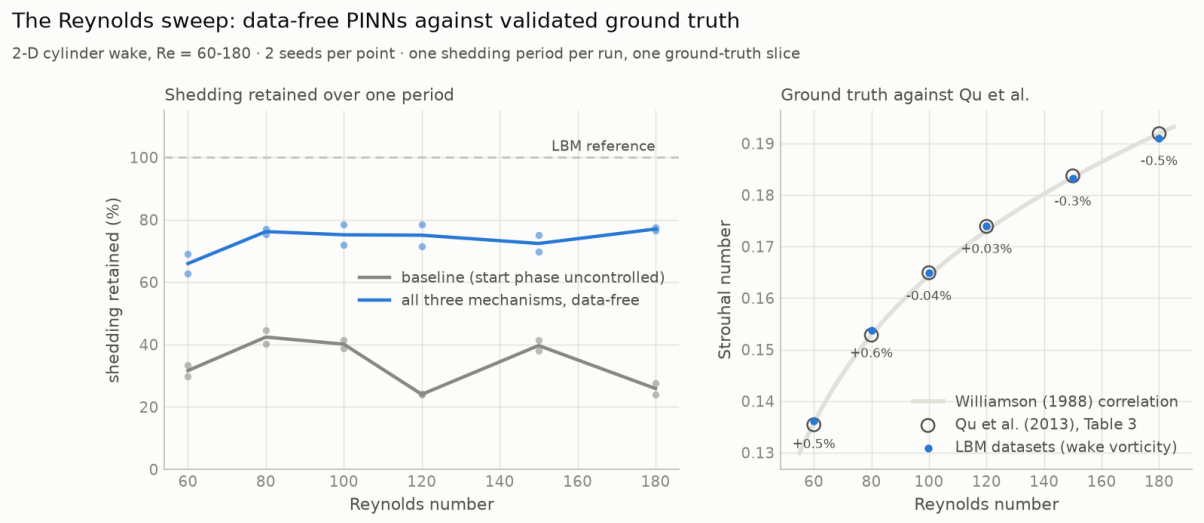


Figure 5: Left: shedding retained against Reynolds number, every seed plotted. Right: each dataset's own validation, measured Strouhal number against Qu et al. (circles) and Williamson's correlation (context).

One start phase per Reynolds number. Each run starts at the beginning of its dataset's record, at whatever phase of the cycle that falls. Section 4 showed that the plain network's retention swings by a factor of two with start phase, so the `baseline` column's variation (24–28% at Re = 120 and 180, 40–45% at Re = 80) should not be given a physical reading. `all` is insensitive to start phase at Re = 100 (between 70% and 77% at each phase), which is why one phase is taken to represent it here; that insensitivity was not re-measured at the other Reynolds numbers.

8 Sequence architectures

Every network so far is a pointwise MLP. PINNsFormer [10] was proposed specifically against PINN failure modes of this kind: it expands each query point (x, y, t) into a short pseudo-sequence in time, $(x, y, t), (x, y, t + \delta t), \ldots$, and mixes the elements with attention. If letting the network see a point's own temporal neighbourhood were enough to stop the collapse, it would be a cheaper fix than marching. Two sequence networks were tested: PINNsFormer at its published size (embedding 32, hidden 512, one encoder and one decoder layer, two heads, $k = 5$, $\delta t = 10^{-4}$, its wavelet activation), and the same embedding and head with a GRU in place of attention, so that attention against recurrence is a controlled comparison. One change

was made to PINNsFormer: the PDE loss is applied at the token for t only rather than at all k pseudo-times. With the published $\delta t = 10^{-4}$ in normalised time the other tokens sit within 10^{-4} of t, so their residuals are near-copies; dropping them makes each network an ordinary function $(x, y, t) \to (u, v, p)$, which runs through the unsteady operator, the marching and the scoring unchanged. The protocol is the one behind the phase-averaged table of Section 5: Re = 100, 20 000 epochs per window, one slice of ground truth, four start phases and two seeds, single-window and marched over four windows. The MLP rows are its seeds 0 and 1, the two the sequence networks were run with, so they differ slightly from the five-seed means of Section 5.

Table 13: Sequence networks against the MLP, data-free, by start phase. Two seeds per cell except PINNsFormer, run once (phase 0, seed 0); the MLP and PINNsFormer runs at that phase and seed are compared in the text. Wall-clock is the median per run on one A100, training and scoring.

	network	parameters	0	1/4	1/2	3/4	mean	wall-clock
single window	MLP	83 459	41.1%	19.3%	42.5%	17.5%	30.1%	6.8 min
	GRU	287 497	33.1%	13.0%	27.8%	13.5%	21.8%	14.1 min
	PINNsFormer	454 619	34.9%	—	—	—	—	256 min
marched	MLP	83 459	72.2%	42.1%	72.4%	40.1%	56.7%	25.9 min
	GRU	287 497	45.4%	19.1%	43.9%	25.7%	33.6%	53.1 min
	PINNsFormer	454 619	77.5%	—	—	—	—	307 min

Neither architecture stops the collapse. On one window the GRU keeps 21.8% of the wake against the MLP's 30.1%, and is below the MLP at every one of the eight phase–seed pairs (by 3.7 to 15.1 points). Its field error is 0.315 against 0.171. PINNsFormer keeps 34.9% where the MLP keeps 40.4% at the same phase and seed. A likely reason neither helps is the pseudo-sequence: at $\delta t = 10^{-4}$ it spans a negligible fraction of the window, so the network sees no more of the time axis than the MLP does.

Marching is still what restores the wake. Paired by phase and seed, marching adds 26.6 points to the MLP and 11.7 to the GRU, and the marched GRU stays 23.2 points below the marched MLP, with the half-cycle swing of Section 4 intact (45% and 44% at phases 0 and 1/2, 19% and 26% at the other two). Marched PINNsFormer is the one result in the other direction: 77.5% against the marched MLP's 69.7% at the same phase and seed, about what all three mechanisms reach (Section 5). It cost 11.8 times the marched MLP's wall-clock, about five hours a run, so the other three phases and the second seed were not run. One run at one phase, beside the marched MLP's swing from 40% to 72% across phases, does not establish an effect; whether attention adds to marching is left open.

9 Physics in a neural operator

The reference paper's FNO is purely data-driven. This section reproduces its recipe and then asks whether physics improves it, and whether the answer belongs to the FNO or to operator forecasting in general.

Reproduction. `underPINN`'s `FNO2D`, trained after Renn et al.: Reynolds-decompose first, map two consecutive frames to the next, apply recursively for ten steps during training with the loss summed over every step; width 80, 24 modes, Adam at 1×10^{-3}, 200 epochs, batch 16. The record is split in time, never interleaved: the first 70% trains, and every forecast starts from a snapshot in the last 30%. Two datasets are used: the production dataset of Section 2, and

the earlier one it replaced, whose fields carry the duct's acoustic mode on top of the wake (Section 2.1).

Contribution. A physics residual is added to that loss. The network predicts (u, v) only, so any residual containing ∇p is unusable; taking the curl of the momentum equations removes the pressure and leaves vorticity transport,

$$\omega_t + u\,\omega_x + v\,\omega_y - \tfrac{1}{\mathrm{Re}}\nabla^2\omega = 0, \qquad \omega = v_x - u_y, \tag{4}$$

evaluated by finite differences on the grid. Continuity is available separately as the cheaper variant. Both residuals are normalised by the residual the LBM data *itself* leaves: the data is weakly compressible and second-order accurate, so driving a prediction below that floor would be fitting discretisation error, not physics.

9.1 On the production dataset

Table 14: Rollout error on the production dataset (relative L_2 on (u, v), held-out initial conditions, two seeds, pairwise mean of adjacent steps). The trained horizon is ten steps, 0.31 shedding periods.

forecast steps	10 (trained)	20	30	50
`data` (paper recipe)	**0.0010**	**0.0010**	**0.0013**	**0.0152**
`physics` (continuity + vorticity)	0.0012	0.0012	0.0017	0.0364
`physics_div` (continuity only)	0.0012	0.0012	0.0018	0.0387

Result. The data-driven reproduction forecasts to an error of 0.0010 at the trained horizon and 0.0152 after 50 steps, about one and a half periods. Neither physics variant improves on it at any horizon: both are 16–18% worse at ten and twenty steps, and 2.4 and 2.5 times as large by fifty (Figure 7, left).

Why this forecast is easy. The error is about ninety times smaller than on the earlier dataset, and the reason limits what the result shows. The production record is a clean limit cycle, so every test initial condition is a state the training portion already passed through at the same phase: held out in time, but close to in-distribution. When the data term alone fits that well, a physics term the data does not itself satisfy exactly (the divergence of Section 2.1) can only pull the prediction away from it. This explanation has not been tested.

9.2 On the earlier dataset, and across backbones

The earlier dataset is the harder test, because its acoustic content makes the held-out states less like the training ones, and it is where the physics term does change the forecast.

Table 15: Rollout error on the earlier dataset (relative L_2 on (u, v), held-out initial conditions, two seeds, pairwise mean of adjacent steps). The trained horizon is ten steps.

forecast steps in shedding periods	10 (trained) 0.31	20 0.61	30 0.92	50 1.53
`data` (paper recipe)	0.093	0.183	**0.296**	**0.365**
`physics` (continuity + vorticity)	**0.074**	**0.132**	0.307	0.772
`physics_div` (continuity only)	0.075	0.140	0.312	0.764

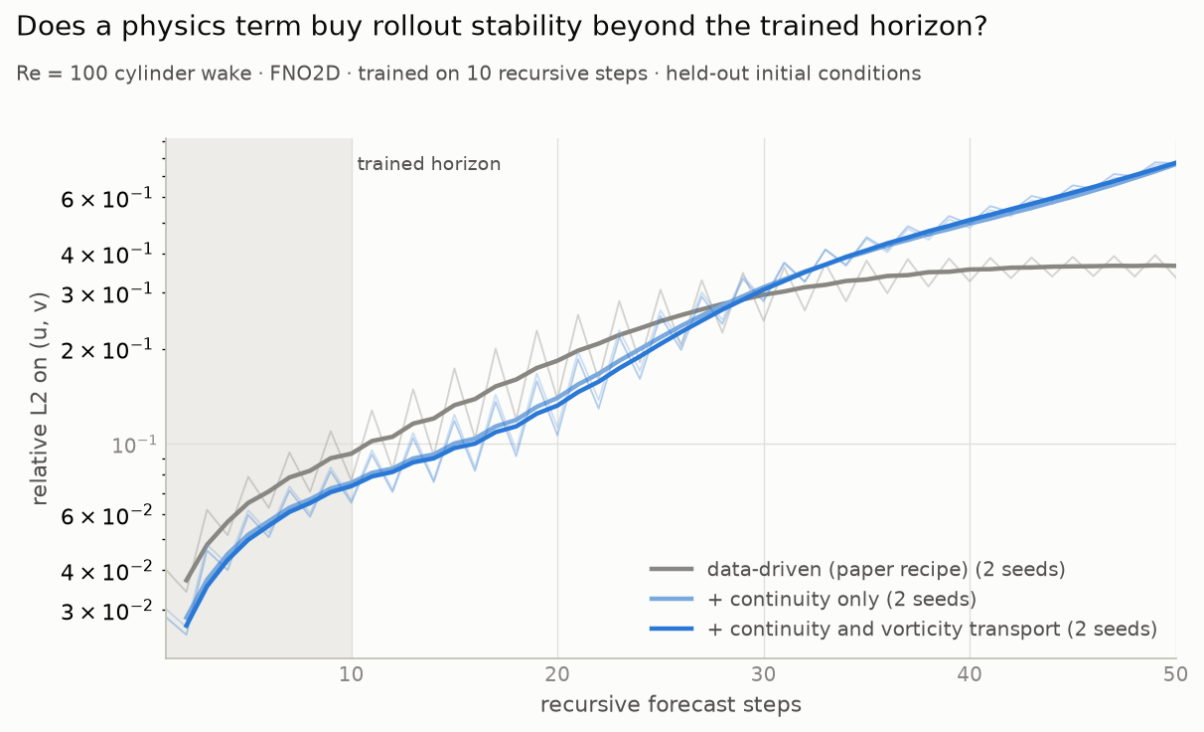


Figure 6: FNO forecast error against horizon on the earlier dataset. Bold lines are pairwise means over adjacent steps; faint lines are raw (see the parity note). The curves cross at approximately one shedding period.

Result. Physics lowers forecast error by 21% at the trained horizon and 28% at twice it, then *crosses over* at step 27–32 across seeds and variants, which is 0.8–1.0 shedding periods. Continuity alone buys nearly all of the benefit, so the effect is not specific to the vorticity term. This is useful, because continuity is much cheaper to compute.

Late horizons. By 50 steps both variants have failed: the error on the fluctuation exceeds unity, meaning that predicting zero fluctuation would beat either rollout. The data-driven curve flattens because it decays towards the mean flow, while the physics-constrained rollouts keep injecting fluctuation energy and overshoot. Across seeds the physics variant reads 0.63 and 0.92 at 50 steps, so the gap is not reproducible. On this dataset, then, physics extends the FNO's useful forecast horizon, but neither model is usable beyond one shedding period.

Even–odd alternation. Rollout error alternates between even and odd steps by 24–44% in *every* variant, physics or not: a period-2 mode of the two-frame autoregressive map. A single step's error can read 30% better or worse purely by which parity it lands on, so all numbers in this section are pairwise means of adjacent steps, and the figure keeps the raw curve visible. On the production dataset the alternation is 4–6%.

Other backbones. The FNO is one of several backbones used for this job. A U-Net [12] (local convolutions in an encoder–decoder with skip connections, which unlike the FNO assumes nothing periodic) and a convolutional GRU in the PhyCRNet pattern [11] (a strided encoder, a ConvGRU cell whose state is carried through the whole rollout, and a decoder) were trained on this dataset under the identical data, loss, rollout and evaluation, data-only and with continuity plus vorticity transport.

Table 16: Three backbones on the earlier dataset: data-driven error, and the change when physics is added (negative is better). Two seeds, pairwise means.

		data-driven error			change with physics		
backbone	parameters	10	20	50	10	20	50
FNO	59.0 M	0.093	0.183	0.365	−21%	−28%	+111%
U-Net	1.93 M	0.056	0.084	0.159	+1%	−1%	−7%
ConvGRU	0.45 M	0.042	0.092	0.220	+30%	+42%	+22%

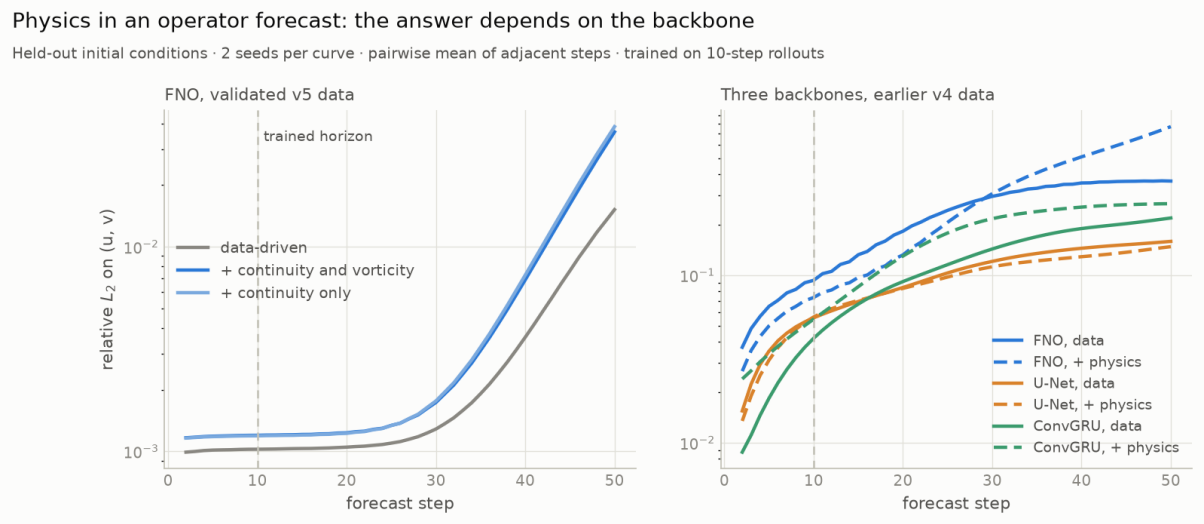


Figure 7: Left: the FNO on the production dataset; physics is worse at every horizon. Right: three backbones on the earlier dataset, data-driven (solid) and with physics (dashed). The dashed line falls below the solid one at short horizons only for the FNO.

Physics helping short and hurting long is a property of the FNO. For the U-Net the physics term changes the error by at most 7% at any of the three horizons, inside the seed spread at fifty steps (0.148 and 0.171 data-driven, 0.133 and 0.163 with physics). For the ConvGRU it is harmful throughout: 30% worse at the trained horizon, on both seeds (0.043 and 0.041 data-driven, 0.055 and 0.055 with physics). Both convolutional networks also forecast better than the FNO without any physics, with 31 and 131 times fewer parameters; at fifty steps the U-Net's error is less than half the FNO's. These comparisons exist on the earlier dataset only; the U-Net and ConvGRU were not rerun on the production one.

Comparison with the reference paper. At the trained horizon the data-driven variant reaches $\epsilon = 0.093$ on the earlier dataset and 0.0010 on the production one, against the paper's $\epsilon < 0.1$. Neither is an improvement on their work, because the data are not comparable: ours is clean, periodic, two-dimensional LBM output at $\mathrm{Re} = 100$, and theirs is noisy experimental PIV at $\mathrm{Re} = 240$–3060 with three-dimensional flow. The comparison shows only that the reproduction works.

10 Conclusions

1. The documented failure reproduces on validated ground truth, and how badly depends on when the run starts: a data-free PINN keeps between 19% and 43% of the wake depending on the start phase, 31.1% on average. Four times the training raises that to 48.6% and leaves the dependence on phase in place.

2. **The three mechanisms are complementary.** Causal time-marching on its own reaches 56.4%, but beats an equal-budget single window by only 7.7 points. Fourier features and the Reynolds decomposition on their own do not help. All three together keep 72.9% (five seeds, ±4.6 points) at every start phase, at a field error of 0.031, and there marching is worth 34.1 points at equal budget.

3. **Two confounds inflate marching results**, and both affected this project's earlier report: giving each window a fresh reference snapshot (about twenty points here), and scoring every configuration at a single start phase.

4. The best data-free configuration matches the phase of the reference to 0.1° over one period, but over five it sheds 10.4% slow and ends about half a cycle behind. Re-anchored with 350 supervised points per period it holds the phase over all five, at about 2% of the data per-window snapshots use.

5. Supervision is needed in small quantity: 100 points per window take retention to 87.4% and 500 to 96.9%, and further data buys accuracy.

6. The ground truth needed two corrections, both of which would affect any lattice Boltzmann cylinder simulation with reflecting lateral walls: an absorbing layer that relaxes toward the free stream, because the domain is an acoustic duct whose transverse mode captures the wake, and an inlet far enough upstream, which mattered more for C_d than the domain height.

7. The result holds across Re = 60–180: 63–79% retained with all three mechanisms, 24–45% without, against ground truth whose St is within 0.6% of Qu et al. at every Reynolds number.

8. **Architecture is not a substitute for causality.** A GRU sequence PINN keeps 21.8% on one window and 33.6% marched, below the plain MLP at every phase either way. A single marched PINNsFormer run reached 77.5%, at twelve times the cost, but one run is not enough to count as a result.

9. **Whether physics helps an operator forecast depends on the data and the backbone.** On the validated data the FNO forecasts to 0.0010 at its trained horizon and physics makes it 17% worse. On the earlier dataset physics improves the FNO by 21% there and crosses over within a period, leaves a U-Net within 7%, and makes a ConvGRU 30% worse, and both convolutional networks beat the FNO without it.

11 Limitations and ongoing work

- **No data-free configuration forecasts** (Section 6). Whether the re-anchoring threshold of 350 supervised points per period moves with the horizon, the Reynolds number or the network is untested, and each re-anchoring schedule was run with two seeds.

- **Soft window continuity.** The marching chain enforces continuity between windows only through the initial-condition loss, which is what produces the small handoff kinks in Figure 4. Roy and Castonguay [13] enforce it exactly through a solution ansatz, on one-dimensional problems; applying it here would remove one source of the accumulated error.

- **Five seeds for the headline, two elsewhere.** The phase-averaged comparison and its equal-budget controls (Sections 4 and 5) have five seeds per start phase, with 95% intervals of ±0.9 to ±1.5 points on the plain and marched networks and ±4.6 on `all`.

The phase-0 ablation, the supervision and Reynolds sweeps, the architecture study and the re-anchoring runs are two seeds each.

- **The equal-budget control has one budget.** Single-window runs were given the marched runs' 80 000 epochs. Whether either kind keeps improving beyond that, and whether marching still wins at a matched budget much larger than both, is untested.
- **The documented case is outside the sweep.** The sweep covers Re = 60–180, its upper limit set by the three-dimensional transition at $\mathrm{Re}_c = 188.5 \pm 1.0$ [16]. The collapse this project responds to was documented at Re = 200 [1], above that transition, so what is reproduced here is the phenomenon rather than their specific case.
- **PINNsFormer was run once** (Section 8). The full grid would take about 66 GPU-hours.
- **The backbone comparison is on the earlier dataset.** The U-Net and ConvGRU were trained only on the data that carries the duct mode, and on the production data the FNO's forecast is close to in-distribution. An operator test whose held-out states are new (a different Reynolds number, or a transient) would separate what each backbone has learned from what it has memorised.
- C_d **is a few per cent high.** Across the sweep it is 3.5–5.7% above Qu et al.; keeping $D = 24$ accounts for about 1.4% of it (Section 2.1), and the rest has not been separated. Nothing in this report depends on C_d: every PINN and operator result is scored against velocity fields and wake probes.

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